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Theme

Towards Trustworthy SBOM Recovery from Compiled Android Applications

An Evidence-Based Component Attribution Framework

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Dedication

[PLACEHOLDER: Personal dedication to be written by the student.]

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Abstract

Software Bills of Materials are increasingly used to support software supply-chain transparency and vulnerability management. However, a generated SBOM is not automatically trustworthy, particularly when it is reconstructed from a compiled Android application without source code or dependency manifests. Compilation, optimization, obfuscation, artifact-family aliases, reference-database gaps, and version similarity can weaken or contradict the evidence supporting a component claim.

This thesis studies trustworthy SBOM recovery as an evidence-based software component attribution problem. It organizes existing Android third-party library and binary component-recovery research into taxonomies of techniques, evidence sources, transformation threats, and evaluation practices. Building on the resulting research gaps, it proposes a conceptual framework in which each recovered claim records component identity, version or version set, evidence provenance, evidence quality, transformation risk, confidence, ambiguity, validation class, and permitted downstream security interpretation.

The implemented APK-to-SBOM PFE system is used as a bounded case study. Its controlled, source-validated, source-constrained, report-only, failed, and limitation evidence demonstrates why library identity, exact version, reference coverage, matcher recall, and vulnerability authority must remain separate. The proposed framework therefore favors auditable partial knowledge and explicit abstention over unsupported certainty, and defines a research agenda for calibrated evidence fusion, provenance-aware SBOMs, realistic benchmarks, and native or dynamically loaded component recovery.

Keywords — software bill of materials, software supply chain, Android, component attribution, provenance, evidence fusion, trustworthiness, uncertainty, vulnerability management.

Résumé

[PLACEHOLDER : La version française sera rédigée après validation de la version anglaise de l'abstract.]

Mots-clés — nomenclature logicielle, chaîne d'approvisionnement logicielle, Android, attribution de composants, provenance, fusion de preuves, confiance, incertitude.

ملخص

سيُستكمل هذا الملخص بعد مراجعة النسخة الإنجليزية واعتماد المصطلحات العلمية المتعلقة بمكونات البرمجيات والأدلة والثقة.

الكلمات المفتاحية — مكونات البرمجيات، سلسلة توريد البرمجيات، أندرويد، الأدلة، مصدر المعلومات، الثقة، عدم اليقين.

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List of Acronyms

Acronym	Definition
APK	Android Package Kit
API	Application Programming Interface
CVE	Common Vulnerabilities and Exposures
DEX	Dalvik Executable
ML	Machine Learning
NVD	National Vulnerability Database
OSV	Open Source Vulnerabilities
PURL	Package URL
R8	Android code shrinker and optimizer
SBOM	Software Bill of Materials
SCA	Software Composition Analysis
SPDX	Software Package Data Exchange
TPL	Third-Party Library

Part I

General Introduction

Chapter 1

General Introduction

1.1 Context

Software supply chains depend on large ecosystems of reusable components. SBOMs improve transparency by recording these components, but many audit scenarios provide only compiled software artifacts. In such cases, an SBOM is reconstructed from indirect and transformed evidence rather than generated from authoritative dependency metadata.

1.2 Problem Statement

A recovered component name or version is not trustworthy merely because a tool emitted it. Trust depends on the origin, independence, transformation resilience, version specificity, and validation status of its evidence. Existing Android detection research provides strong technical signals, but lacks a unified model connecting those signals to claim provenance, uncertainty, and permitted vulnerability interpretation.

1.3 Motivation

1.4 Research Questions

1. Which evidence sources are available for recovering software components from compiled Android applications?
2. How should evidence strength for identity and exact-version attribution be assessed?
3. How should conflicting, missing, transformed, or source-dependent evidence be represented?
4. What information must accompany a reconstructed SBOM claim for it to be auditable and useful in vulnerability management?

1.5 Objectives and Contributions

The thesis contributes a taxonomy of recovery techniques, an evidence and transformation taxonomy, an evidence-based claim model, a validation-lane model, a vulnerability-authority policy, and a research agenda illustrated through selected evidence from the associated PFE project.

1.6 Scope

1.7 Thesis Organization

Part II

Background and Research Methodology

Chapter 2

Background and Research Methodology

2.1 Introduction

This chapter establishes the concepts and review method used to reason about trustworthy recovery of software components from compiled Android applications. The thesis treats a recovered SBOM as a set of evidence-backed claims rather than as a simple list of names and versions. This distinction requires a common vocabulary for component identity, provenance, uncertainty, and transformation risk, followed by a transparent method for selecting and comparing prior research.

2.2 Software Supply Chains and SBOMs

Modern applications assemble code, resources, native binaries, generated artifacts, and transitive dependencies from many independently maintained projects. An SBOM records these components so that operators can reason about inventory, licensing, provenance, and known vulnerabilities. Standards define how such information is represented, but a representation format does not establish that the recorded component is actually present or that its exact version is known.

The framework considers both SPDX and CycloneDX, while using Package URL as a standard package-identity model [1, 2, 3]. These standards therefore define the target vocabulary and exchange structures, while the attribution framework developed in this thesis defines the evidential conditions under which individual fields may be populated.

2.2.1 Authoritative and Reconstructed SBOMs

An authoritative build-time SBOM can use dependency resolution metadata, lockfiles, build graphs, and package-manager records. By contrast, an artifact-reconstructed SBOM begins after those authoritative records have been removed or made unavailable. It must infer a component from observable residue in an APK, compare that residue with references, and preserve the uncertainty introduced by compilation and incomplete coverage.

The distinction is central to this thesis. A reconstructed component identity may

be credible even when its exact version is unresolved. Similarly, a syntactically valid CycloneDX or SPDX document may contain semantically weak claims. Trustworthiness is consequently a property of each claim and its evidence trail, not of the file format alone.

2.3 Trust, Evidence, Provenance, and Uncertainty

2.3.1 Evidence and Claims

An evidence item is an observable fact used to support or challenge a component claim. Examples include a package fragment, an opcode fingerprint, a class-dependency relation, a resource marker, a native-library name, or a source-validated expected dependency. A claim is the interpretation built from one or more such items, for example that a library family is present or that a particular version is the best-supported candidate.

2.3.2 Provenance and Independence

Provenance records where evidence came from, how it was extracted, and which transformations may have affected it. Two matching fields are not independent when they are derived from the same underlying artifact feature. Confidence must therefore not be increased merely by counting correlated signals as separate confirmations.

2.3.3 Uncertainty and Abstention

Uncertainty can arise from missing evidence, conflicting evidence, incomplete reference coverage, adjacent versions with near-identical code, partial library inclusion, or destructive optimization. The framework allows ambiguity sets and abstention because an explicit unresolved result is more auditable than a forced exact version. This policy also limits downstream vulnerability authority: a vulnerability finding cannot be stronger than the identity and version evidence on which it depends.

2.4 Android Compiled-Artifact Context

Android compilation and R8 transformations directly affect the stability and interpretability of recovered evidence [4].

Java or Kotlin source is compiled and transformed into DEX bytecode, while resources and manifests are packaged into an APK. Dependency boundaries that were explicit during the build may disappear when classes are merged into one or more DEX files. R8 can rename identifiers, remove unused code, rewrite control flow, merge classes, inline methods, and change call sites. Packaging may also introduce multidex layouts, shaded or relocated code, native libraries, generated classes, and dynamically loaded components.

These transformations do not affect all evidence equally. Names and package trees are inexpensive but fragile. Opcode, control-flow, call, and dependency structures can survive renaming, yet they are still altered by shrinking and optimization. Resources and metadata can provide independent support, but they may identify only a family rather

than an exact version. The state-of-the-art taxonomy in chapter 3 therefore classifies both evidence sources and the transformations that threaten them.

2.5 Software Component Attribution

2.5.1 Identity, Version, and Security Authority

Component attribution contains at least three distinct decisions:

1. **identity attribution**: whether a component family or coordinate is present;
2. **version attribution**: which exact version or bounded version set is supported; and
3. **security interpretation**: which vulnerability statements are permitted by the preceding evidence.

A method that performs well at library-family detection does not automatically provide exact-version authority. Likewise, a vulnerability database match is not confirmation that the affected version is present. Keeping these decisions separate prevents a plausible identity match from being silently promoted into an exact and security-critical statement.

2.5.2 Reference Dependence

Many attribution methods compare an application with profiles extracted from known library releases. Such a method can recognize only what its reference corpus represents. Failure may therefore result from the matcher, the extractor, or a missing reference rather than from absence of the component. Reference coverage is treated as a first-class validity condition throughout the thesis.

2.6 Evidence Sources in APKs

The review extracts evidence into recurring families:

- lexical and namespace evidence, including package and class names;
- bytecode and semantic evidence, including opcodes, API references, literals, and method descriptors;
- structural evidence, including control-flow, call, inheritance, and class-dependency graphs;
- resource and packaging evidence, including manifests, resource names, assets, and native-library files;
- external reference and metadata evidence, including library profiles, Maven coordinates, dependency relations, and release chronology;
- learned representations derived from bytecode or structural features;

- validation evidence, including controlled builds, source manifests, and source-constrained observations.

No family is assumed to be universally authoritative. The framework evaluates its origin, transformation resilience, version specificity, independence, coverage, and validation status.

2.7 Structured Literature Review Methodology

The literature study follows a frozen structured mapping protocol adapted from software-engineering systematic-review guidance [5]. It is not presented as a completed exhaustive systematic literature review because the initial corpus was assembled iteratively before the protocol freeze. Instead, the protocol makes the existing corpus auditable and governs all future search refreshes, screening, extraction, and synthesis.

2.7.1 Review Questions and Search Strategy

The review asks which evidence sources and attribution techniques are used, which Android transformations threaten them, how identity and version results are evaluated, how uncertainty and provenance are represented, and which gaps prevent trustworthy SBOM claims. Searches combine four concept blocks: Android artifacts; third-party components or SBOMs; detection or attribution tasks; and obfuscation, optimization, version, or vulnerability terms.

Primary sources include ACM Digital Library, IEEE Xplore, SpringerLink, ScienceDirect, USENIX, and arXiv. Google Scholar is restricted to discovery and citation chaining. The date scope begins in 2015 and continues through the recorded search date. Full query text and database adaptations are preserved in chapter A.

2.7.2 Eligibility and Screening

Eligible studies propose or evaluate Android binary component attribution, version identification, relevant benchmarks, empirical assessments, or SBOM-oriented recovery frameworks. Studies based only on authoritative source manifests, unrelated malware classification, product pages, duplicates, and records without extractable methodology are excluded.

Screening proceeds through deduplication, title-and-abstract screening, full-text assessment, and evidence-tier assignment. Related conference and journal publications are treated as one study family unless the later publication contributes distinct methods, datasets, or results.

2.7.3 Evidence Tiers

The frozen seed corpus contains 20 publication units. Thirteen have locally available full texts and form *Tier A*. Seven are *Tier B* supplementary records whose metadata or structured notes are available but whose local full texts have not yet been verified. Tier B records may inform search coverage and provisional gap mapping, but they cannot

independently support precise quantitative conclusions. Standards and official platform documentation form contextual *Tier C* and are not counted as primary studies.

2.8 Analysis Dimensions

Each included study is extracted across common dimensions: task granularity, input assumptions, segmentation, evidence families, matching or learning method, transformation coverage, ground truth, dataset composition, metrics, confidence and ambiguity behavior, reference dependence, artifact availability, SBOM output, vulnerability mapping, and stated limitations.

Tier A studies are additionally assessed against eight reporting-quality questions covering objective clarity, reproducibility, ground truth, metric definition, transformation conditions, identity/version separation, artifact availability, and threats to validity. Quality affects the weight of a study in synthesis but does not automatically erase historically important work.

Synthesis is narrative and taxonomy-based. Metrics are compared only when their task unit, dataset, ground truth, and transformation conditions are compatible. Original-paper results remain separate from later independent benchmarks, and library-level, class-level, and version-level results are never merged into a single ranking.

2.9 Threats to the Review

2.9.1 Search and Selection Bias

The pre-protocol corpus was built iteratively and may overrepresent well-known Android security venues or studies connected through citation chains. The frozen search blocks, multiple publisher databases, and explicit refresh log reduce but do not eliminate this risk.

2.9.2 Availability and Recency

Several recent or emerging studies currently exist only as supplementary records. Treating them as Tier B prevents incomplete metadata from acquiring the same authority as inspected full text. Their classifications may change after verification, and all such changes must be logged.

2.9.3 Heterogeneous Evaluation

Reported accuracy values often refer to different task definitions, datasets, library versions, obfuscators, and ground-truth procedures. Direct numerical ranking can therefore be misleading. The review preserves these conditions and prefers qualified comparison over pooled scores.

2.9.4 Reviewer and Extraction Bias

The current review is primarily maintained by one researcher. A fixed extraction schema, explicit `NR` values for unreported information, source citations, and retained structured records provide an audit trail. A later independent screening sample would further strengthen reliability.

2.10 Conclusion

This chapter separated a valid SBOM representation from the evidential strength of its contents and defined component attribution as a sequence of identity, version, and security-authority decisions. It also froze a structured mapping protocol that prevents unavailable full texts, incompatible metrics, and correlated signals from being silently promoted into strong claims. The next chapter applies these rules to build the state-of-the-art taxonomies that motivate the proposed trust framework.

Part III

State of the Art and Taxonomy

Chapter 3

State of the Art and Taxonomy

3.1 Introduction

3.2 Taxonomy of Component-Recovery Techniques

3.3 Taxonomy of Evidence Sources

3.4 Taxonomy of Transformation Threats

3.5 Representative Android TPL Detection Approaches

Reference-profile systems such as LibScout provide an important baseline for understanding identity and version attribution from compiled Android code [6].

3.6 Evaluation Practices and Ground Truth

3.7 Comparative Analysis

3.8 Limitations of Existing Research

3.9 Research Gap

3.10 Conclusion

Part IV

Proposed Framework

Chapter 4

Proposed Evidence-Based Trust Framework

4.1 Introduction

The proposed framework treats every recovered SBOM entry as a claim that must be accompanied by evidence, provenance, uncertainty, and an explicit boundary on how the claim may be used.

4.2 Framework Design Principles

1. Every claim exposes its supporting evidence.
2. Library identity and exact version are separate claims.
3. Missing evidence is not interpreted as component absence.
4. Conflicting evidence remains visible.
5. Confidence is bounded by transformation and reference-coverage risks.
6. Vulnerability authority cannot exceed component and version authority.
7. Source-assisted validation is distinguished from APK-only inference.
8. Report-only assistance is distinguished from authoritative output.

4.3 Recovered Claim Model

A recovered claim contains the analyzed APK, component identity candidate, artifact family, version or version set, evidence list, evidence provenance, coverage context, transformation risk, confidence class, ambiguity, validation class, and permitted downstream interpretation.

4.4 Evidence Hierarchy

Table 4.1: Initial evidence hierarchy for reconstructed component claims.

Level	Meaning	Permitted interpretation
E0	No component evidence	No package-level claim
E1	Weak contextual evidence	Investigation hint only
E2	Identity-supporting evidence	Candidate component identity
E3	Independent multi-source identity evidence	Strong identity claim
E4	Version-discriminating APK evidence	Bounded exact or version-set claim
E5	Externally validated exact evidence	Exact claim in the validation context

4.5 Evidence Quality Dimensions

4.6 Evidence Fusion, Conflict, and Abstention

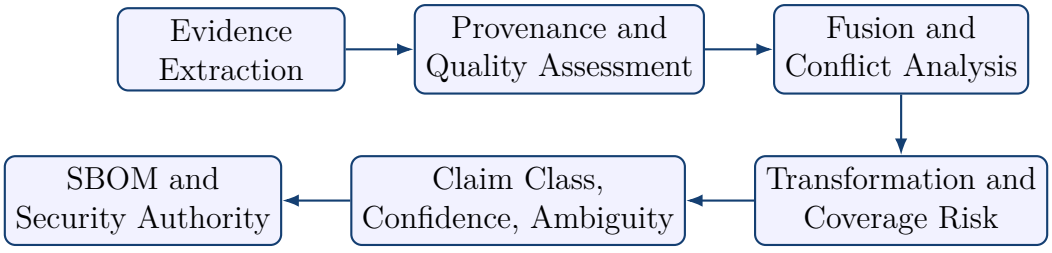


Figure 4.1: Conceptual flow of the proposed evidence-based trust framework.

4.7 Claim Classes

4.8 Validation and Evidence Lanes

4.9 Vulnerability-Claim Policy

4.10 Recovered Claim Lifecycle

4.11 PFE System as a Case Study

Selected PFE observations will be used to demonstrate controlled exact recovery, clean and partial real-APK evidence, version-band ambiguity, reference coverage versus matcher recall, source-known upper-bound analysis, failed promotions, report-only AI, and native or hidden-code boundaries.

4.12 Framework Evaluation

4.13 Limitations and Research Agenda

4.14 Conclusion

Part V

General Conclusion

Chapter 5

General Conclusion and Future Directions

5.1 General Conclusion

5.2 Answers to the Research Questions

5.3 Main Contributions

5.4 Limitations

5.5 Future Research

5.6 Closing Remarks

Appendix A

Literature Review Protocol

This appendix records version 1.0 of the structured literature mapping protocol, frozen on 21 June 2026. It is the operational companion to the methodology summarized in chapter 2. The review follows the planning, selection, quality-assessment, extraction, and synthesis principles of software-engineering review guidance [5], while explicitly acknowledging that the seed corpus was assembled iteratively before the protocol was frozen.

A.1 Research Questions

RQ-R1 Which evidence sources are used to identify Android third-party components from compiled applications?

RQ-R2 Which techniques support library identity, exact-version attribution, or both?

RQ-R3 Which compilation, optimization, obfuscation, packaging, and dependency transformations threaten each evidence source?

RQ-R4 How do studies construct ground truth and evaluate library-level, version-level, and vulnerability-related claims?

RQ-R5 How do studies represent confidence, ambiguity, provenance, abstention, partial inclusion, and conflicting evidence?

RQ-R6 Which research gaps prevent recovered component claims from becoming trustworthy and auditable SBOM statements?

A.2 Search Strategy

A.2.1 Sources and Scope

The primary search sources are ACM Digital Library, IEEE Xplore, SpringerLink, ScienceDirect, USENIX, and arXiv. Google Scholar is used for discovery and backward or forward citation chaining rather than as the sole bibliographic authority. The publication window begins in 2015 and ends on the date recorded for each search refresh. English full text is preferred; English title and abstract are mandatory for screening.

A.2.2 Concept Blocks

Table A.1: Search concept blocks

Block	Terms
Platform	Android, APK, DEX, Dalvik
Component	third-party library, TPL, software component, dependency, SBOM, software bill of materials
Task	detect*, identif*, recogn*, recover*, attribut*, fingerprint*, match*
Threat or specificity	obfuscat*, R8, ProGuard, optim*, shrink*, inlin*, version*, vulnerab*

The canonical query is:

```
(Android OR APK OR DEX OR Dalvik)
AND ("third-party library" OR TPL OR "software component" OR dependency
   OR SBOM OR "software bill of materials")
AND (detect* OR identif* OR recogn* OR recover* OR attribut*
   OR fingerprint* OR match*)
AND (obfuscat* OR R8 OR ProGuard OR optim* OR shrink* OR inlin*
   OR version* OR vulnerab*)
```

Database-specific syntax may change, but a refresh must preserve the four concept blocks and record the exact executed query, date, raw result count, deduplicated count, and selection decisions.

A.3 Inclusion and Exclusion Criteria

A.3.1 Inclusion

A publication is included when it:

1. addresses component, library, package, or version recovery from Android binaries or evaluates tools that do;
2. studies evidence obtainable from compiled artifacts, reference libraries, package metadata, or related dependency knowledge;
3. contributes a method, benchmark, empirical assessment, taxonomy, or SBOM-oriented framework relevant to compiled-artifact attribution;
4. permits extraction of at least the task, evidence source, evaluation basis, and limitations; and
5. has an accessible English title and abstract.

A.3.2 Exclusion

A publication is excluded when it:

1. relies only on source dependency manifests and does not study binary recovery;
2. concerns malware classification, clone detection, or vulnerability scanning without a component-attribution contribution;
3. is a tutorial, slide deck, blog post, product page, or duplicate;
4. reports no separable Android result;
5. lacks enough information to identify its method or evaluation basis; or
6. is a shorter version of an included extended study and contributes no distinct method, dataset, or result.

A.3.3 Screening and Evidence Tiers

Screening proceeds through deduplication, title-and-abstract screening, full-text screening, and tier assignment. The evidence tiers are:

Tier A — full-text verified A local full text has been inspected and the publication may support structured and quantitative extraction.

Tier B — supplementary record Metadata or a structured note exists, but the local full text has not yet been verified. The record may guide search and provisional gap mapping, but not precise quantitative claims.

Tier C — contextual authority Standards, official documentation, and project artifacts used for definitions or framework design. They are not counted as primary studies.

Excluded The reason for exclusion is recorded and the item is not used in synthesis.

A.4 Data Extraction Schema

Table A.2: Structured extraction fields

Dimension	Extracted information
Bibliographic identity	Title, authors, year, venue, DOI or stable URL, publication type, study family, evidence tier.
Objective and task	Research objective; family, library, class, version, SBOM, or vulnerability granularity.
Inputs and assumptions	Required APK artifacts, reference corpus, metadata, training data, and coverage assumptions.
Segmentation	Package boundaries, class hierarchy, dependency graph, clustering, or other module-decoupling method.

Dimension	Extracted information
Evidence	Names, APIs, opcodes, control flow, call/dependency graphs, literals, resources, native code, metadata, learned representations, and cross-component relations.
Decision method	Hashing, similarity, graph matching, ranking, classification, probability, learned model, or evidence fusion.
Uncertainty behavior	Threshold, confidence, competing versions, abstention, partial matching, provenance, and conflict handling.
Transformations	Renaming, package flattening, shrinking, optimization, control-flow rewriting, method inlining, repackaging, multidex, partial inclusion, native code, and dynamic loading.
Evaluation	Dataset, ground truth, evaluation unit, metrics, baselines, obfuscation setting, and independent benchmark results.
Operational support	Artifact availability, runtime, scalability, SBOM format, vulnerability mapping, and transitive-dependency handling.
Limitations and relevance	Threats to validity, negative findings, and adoption, adaptation, or rejection relevance to the framework.

Unreported information is recorded as `NR`; it is not inferred from neighboring fields or from another publication in the study family.

A.5 Quality Assessment

Tier A publications receive a score of 1, 0.5, or 0 for each question:

1. Is the research objective clear?
2. Is the method described well enough to understand or reproduce?
3. Are the dataset and ground truth described?
4. Are the task unit and metrics explicit?
5. Are transformation or obfuscation conditions stated?
6. Are identity and exact-version claims separated?
7. Is an implementation, dataset, or artifact available?
8. Are limitations or threats to validity acknowledged?

The score controls synthesis weight rather than automatic inclusion. Historically important studies may remain in the map with an explicit reporting-quality limitation.

A.6 Synthesis Rules

- Narrative and taxonomy-based synthesis is the default.
- Incompatible datasets and metrics are not pooled into a common leaderboard.

- Library-level, class-level, and version-level results remain separate.
- Original-paper results remain separate from independent benchmark results.
- Obfuscation and optimization conditions must accompany performance claims.
- Artifact availability is assessed independently of reported accuracy.
- Tier B records cannot independently support precise numerical claims.
- Identity authority, version authority, and vulnerability authority are separate.
- Explicit ambiguity or abstention is preferred to unsupported exactness.

A.7 Selected Studies

Table A.3: Frozen seed corpus and current evidence status

ID	Publication or study label	Year	Status
S01	LibScout [6]	2016	Tier A
S02	LibD [7]	2017	Tier A
S03	LibPecker [8]	2018	Tier A
S04	ORLIS [9]	2018	Tier A
S05	LibID [10]	2019	Tier A
S06	Automated Third-Party Library Detection: Are We There Yet? [11]	2020	Tier A
S07	Systematic Assessment of Android TPL Detection Tools [12]	2021	Tier A
S08	ATVHUNTER [13]	2021	Tier A
S09	LibScan [14]	2023	Tier A
S10	Empirical Study of TPL Detection in iOS and Android [15]	2024	Tier A
S11	LibHunter / Code-Optimization Impact Study [16]	2024	Tier A
S12	LibAttention [17]	2025	Tier A
S13	SAD [18]	2025	Tier A
S14	LibRadar	2016	Tier B
S15	Libra	2023	Tier B
S16	LibMD	2025	Tier B
S17	TPL-Benchmark / Revisiting TPL Detection	2025	Tier B
S18	UniBOM	2025	Tier B
S19	LibScope / Towards Global Matches	2026	Tier B
S20	ObfDetector / Advanced-Obfuscation Robustness Study	2026	Tier B

The seed corpus contains 20 publication units: 13 Tier A full texts and seven Tier B supplementary records. This count is a protocol checkpoint, not a claim that all 20 have equal evidential authority. A Tier B record is promoted only after full-text verification, bibliographic checking, structured extraction, and quality assessment.

A.8 Audit and Change Control

The current extraction records are maintained under `papers/data/`. Each search refresh must append the date, sources, query variants, raw and deduplicated result counts, inclusion and exclusion decisions, tier changes, and taxonomy effects to the master-thesis session log. Eligibility, evidence tiers, and synthesis rules may be amended only through a dated protocol change note; silent changes are prohibited.

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